Major Project Report: Cost of Living Analysis - City overview

Course: Data Analytics and Visualization

Student Name:Kanade Saish Santosh

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1. Project Objective

The objective of this project is to apply a complete data analytics workflow to evaluate urban affordability and living standards across 150 Indian cities. Using web-scraped data from Numbeo, the project aims to identify the socio-economic drivers of living costs and predict financial capacity (salary) based on regional price indices.

Specifically, this study segments cities into economic clusters using K-Means Clustering and evaluates the predictive power of various cost-of-living metrics on net income using Random Forest Regression, providing insights into the real-world trade-offs between income and expenses.

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2. Methodology & Data Collection

- **Data Source: Primary data was web-scraped from the Numbeo website, focusing on consumer price indices.**
- **Dataset Size: 150 unique urban centers (observations) with 57 initial economic indicators.**
- **Key Features: The analysis utilized a wide array of indicators, categorized into essential spending groups.**

Variable Name	Data Type	Description & Significance

City	Categorical	The metropolitan area analyzed (e.g., Mumbai, Bangalore, Delhi).
Market Basket Index	Numeric	Engineered feature: Total cost of 11 essential food staples (milk, rice, eggs, etc.).
Transportation Index	Numeric	Engineered feature: Costs of local transit, fuel, and taxi tariffs.
Housing (1 BHK)	Numeric	Monthly rent for a one-bedroom apartment in the city center.
Net Salary	Numeric	Target variable: Average monthly net salary after tax.
Utilities	Numeric	Basic costs (electricity, water, etc.) for an 85m^2 apartment.
Childcare Index	Numeric	Engineered feature: Monthly preschool fees and normalized school tuition.

3. Data Cleaning and Preparation

A rigorous preprocessing pipeline was implemented to handle the high dimensionality and noise of the scraped data:

1. **Standardization of Features: Column names were programmatically cleaned to a snake_case format (e.g., `average_monthly_net_salary_after_tax`) to ensure compatibility with Python libraries.**
2. **Imputation of Missing Values: As some smaller cities had missing price points, `SimpleImputer` was used to fill numerical gaps with the median value of the column, ensuring the dataset remained statistically robust.**
3. **Feature Engineering (Indices): To simplify the analysis, 57 raw features were aggregated into four primary indices: `market_basket_index`, `transportation_cost_index`, `sports_leisure_index`, and `childcare_cost_index`.**
4. **Scaling: For clustering purposes, features were scaled using `StandardScaler` to bring variables like "Rent" and "Coffee Price" into a comparable range ($mean=0, std=1$).**

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4. Exploratory Data Analysis (EDA)

EDA was conducted to visualize the distribution of wealth and the cost of living across the country.

Key Analytical Insights:

- **Salary Skewness: The majority of Indian cities offer net salaries between ₹30,000 and ₹50,000, with high-paying outliers like Hyderabad and Pune reaching above ₹110,000.**
- **Housing as a Driver: A strong positive correlation was observed between city-center rent and monthly salary, suggesting that high-income cities disproportionately tax residents through housing.**

- Utility Uniformity: Unlike housing, utility costs showed a narrow distribution, indicating that electricity and water are relatively fixed costs regardless of a city's average income.
- Tipping Point: Analysis of the "Salary-to-Rent" ratio indicated that while "Tier-1" cities have the highest salaries, their actual affordability is lower than "Tier-2" cities due to hyper-inflated rent prices.

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5. Insights from Data Visualization

1. Top Salary Hubs: Bar charts identified Hyderabad as the leader in net monthly income, followed closely by Pune and Bangalore.
2. Rental Disparities: Scatter plots of "Rent vs. Salary" highlighted that Mumbai's rent is an extreme outlier, drastically impacting the living standards compared to cities like Jaipur or Ahmedabad.
3. Food & Transit Trends: The pairplot revealed that "Market Basket" (groceries) and "Transportation" costs are more uniform across cities than housing prices, making them less significant as differentiators for city tiers.
4. Correlation Heatmap: The highest correlation coefficients were found between Salary and 1-Bedroom Rent, validating housing as the most critical economic predictor in the dataset.

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6. Model Building & Performance

Two distinct approaches were taken to model the data:

A. Supervised Learning (Salary Prediction)

- Linear Regression: Achieved poor performance ($R^2 < 0$), proving that the relationship between urban costs and income is complex and non-linear.

- **Random Forest Regressor: After hyperparameter optimization, the model achieved a high R^2 of 0.80.**
- **Feature Importance: The model identified Housing Costs and the Market Basket Index as the most influential factors in predicting a city's average salary.**

B. Unsupervised Learning (City Clustering)

- **K-Means Clustering: Based on the Elbow Method, the cities were divided into 3 distinct clusters:**
  - **Cluster 0 (High-Cost Hubs): Major metros with high salaries and high rents.**
  - **Cluster 1 (Affordable Tier-2): Cities with moderate costs and balanced salaries.**
  - **Cluster 2 (Low-Cost Regions): Areas with significantly lower living costs but also lower income potential.**

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7. Conclusion

This analysis successfully maps the economic landscape of 150 cities using data-driven methodologies. The project proves that salary levels are highly dependent on the cost of housing and local property markets. While major metros offer higher pay, the Cluster Analysis suggests that the "sweet spot" for urban dwellers often lies in the moderate-cost "Tier-2" cities where the ratio of income to essential expenses is most favorable.